Second-Year Project Literature Survey

Graph Neural Network-Based Dynamic User Pairing and Slot Assignment for Hybrid NOMA Networks

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Contents

Executive Summary 2

1 Introduction and Related Research Themes 2

2 Implementation Analysis of the Inspected Project 2

3 Individual Paper Summaries 3

4 Comparative Synthesis 6

5 Research Gaps and Project Motivation 6

Executive Summary

This report contextualizes power-domain Non-Orthogonal Multiple Access (PD-NOMA) user pairing, fairness, and Successive Interference Cancellation (SIC) within graph-based wireless resource allocation, link scheduling, and Sparse Code Multiple Access (SCMA) message passing[cite: 2, 7, 10]. The inspected implementation consists of a synthetic single-cell simulator utilizing an oracle-supervised edge-conditioned Graph Neural Network (GNN), deterministic greedy constraint enforcement, two-user PD-NOMA with residual SIC, and a standalone SCMA detector module[cite: 1, 7, 10].

The checked-in benchmark suite serves as an implementation-specific comparative evaluation rather than a generalized field proof of superiority. Notably, the hybrid scheduler employs a closed-form rate proxy for SCMA mode selection, whereas the full SCMA codebook and Message Passing Algorithm (MPA) detection engine are evaluated independently[cite: 10, 11]. While published academic literature motivates learned graph policies and clarifies underlying physical layer radio-access mechanisms[cite: 1, 7, 8], it does not establish that this exact composite system is novel. Consequently, this report emphasizes concrete model validation and identification of research gaps.

1. Introduction and Related Research Themes

Power-Domain Non-Orthogonal Multiple Access (PD-NOMA) superposes multiple user signals over shared time-frequency resource blocks, leveraging receiver-side Successive Interference Cancellation (SIC) to decode messages[cite: 1, 2]. Achievable rate gains are highly dependent on channel gain disparity, dynamic power allocation, decoding sequence, and residual SIC error[cite: 3, 4, 6]. Consequently, user pairing is an essential optimization step: suboptimal pairing undermines sum-rate gains, whereas strict fairness-aware power allocation can conflict with overall throughput maximization[cite: 3, 4, 5, 6]. In parallel, SCMA presents a code-domain non-orthogonal access scheme where multidimensional sparse codebooks construct a factor graph, allowing low-complexity iterative detection via the Message Passing Algorithm (MPA)[cite: 10, 11].

Wireless resource allocation and link scheduling problems are inherently combinatorial. Graph Neural Networks (GNNs) provide permutation-equivariant representations for nodes (users) and directed edges (interference/channel links)[cite: 7, 8, 9]. Prior research demonstrates graph-based power control and link scheduling designed for scalable network topologies[cite: 7, 8, 9]. These works validate the architectural selection of graph representations, though they do not automatically certify every specific engineering choice in this project's NOMA pairing implementation.

This survey examines three primary research connections:

(i) Conventional PD-NOMA pairing algorithms, fairness constraints, and imperfect SIC modeling[cite: 1, 2, 3, 4, 6];

(ii) Graph learning methodologies applied to wireless link scheduling and optimization[cite: 7, 8, 9]; and

(iii) SCMA factor-graph message-passing detection[cite: 10, 11].

2. Implementation Analysis of the Inspected Project

A detailed code and repository audit was conducted to confirm the exact radio and algorithmic mechanisms implemented:

- **Data and Channel Model:** The primary path uses synthetic network generation. The default topology represents a single downlink cell with 10 users and 5 resource slots distributed across a 20–500 m annulus. Channel modeling includes 2.4 GHz carrier frequency, 10 MHz total bandwidth, 1 W transmit power, log-distance path loss (exponent 3.5), and Rayleigh fading. Graph training data defaults to 200 training, 40 validation, and 50 test scenarios.
- **Graph Architecture and Training:** A fully connected directed graph without self-loops is constructed. Nodes feature 6 attributes, while directed edges contain 4 attributes. A native PyTorch edge-conditioned message-passing network (3 layers, 64 hidden units) predicts pair-affinity logits and per-user slot preferences. Training is supervised using oracle targets via weighted binary cross-entropy plus 0.3× slot cross-entropy. The oracle uses exhaustive partition enumeration for ≤ 10 users and maximum-weight matching surrogates for larger networks.
- **Scheduling Post-Processing and PHY Layer:** Affinity logits are mapped to disjoint pairs via deterministic greedy matching. Downlink PD-NOMA employs a static 0.75/0.25 weak/strong user power split with configurable residual SIC (default coefficient 0.01).
- **Saved Benchmark Performance:** On the default 10-user/5-slot test benchmark (50 test scenarios), reported mean sum-rates are: GNN (17.686 ± 6.832 Mbps), Oracle (17.594 ± 7.015 Mbps), and OMA (19.493 ± 8.276 Mbps). Corresponding fairness indices and outage rates are 0.529/10.0% (GNN), 0.523/6.4% (Oracle), and 0.335/50.4% (OMA).
- **Implementation Caveats Identified:**
  1. *SCMA Integration Gap:* The scheduler's mode selection utilizes a closed-form rate proxy; the 6-layer/4-resource SCMA codebook and iterative MPA detector are executed in an independent SNR sweep script[cite: 10, 11].
  2. *Edge Feature Tautology:* The edge feature labeled 'channel correlation' is computed as the normalized magnitude of complex scalar product ($\frac{|h_i h_j^*|}{|h_i||h_j|}$). For non-zero complex scalars, this evaluates identically to 1.0, providing no spatial channel correlation variance.

3. Individual Paper Summaries

[1] Saito et al. (2013) – Non-Orthogonal Multiple Access for Cellular FRA

- **Problem & Method:** Introduces downlink power-domain NOMA for cellular access, multiplexing users in the power domain over identical radio resources with receiver-side SIC[cite: 1].
- **Findings:** Demonstrates superior system throughput and cell-edge user experience compared to conventional orthogonal access[cite: 1].
- **Limitations & Project Relevance:** Foundational PHY proposal; does not consider learned graph scheduling or multi-slot constraints. Groundwork for the simulator's 2-user superposition/SIC equations[cite: 1].

[2] Ding et al. (2017) – A Survey on NOMA for 5G Networks

- **Problem & Method:** Comprehensive survey organizing power-domain NOMA, code-domain NOMA, receiver complexity, user pairing, and resource allocation[cite: 2].
- **Findings:** Synthesizes key theoretical capacity bounds and practical implementation challenges[cite: 2].

- **Limitations & Project Relevance:** Predates modern deep graph learning literature. Establishes the conceptual baseline for pairing and fairness metrics[cite: 2].

[3] Zhu et al. (2019) – Optimal User Pairing for Downlink NOMA

- **Problem & Method:** Derives mathematical conditions for optimal 2-user pairing in downlink NOMA under dynamic power allocation[cite: 3].
- **Findings:** Proves that pairing policy significantly alters achievable rate regions relative to random or purely orthogonal strategies[cite: 3].
- **Limitations & Project Relevance:** Analytical optimization focused purely on NOMA; does not incorporate GNN representations or hybrid mode switching[cite: 3]. Direct precedent for near-far and oracle pairing baselines.

[4] Ding, Fan, and Poor (2016) – Impact of User Pairing on 5G NOMA

- **Problem & Method:** Analyzes performance bounds of fixed-NOMA and cognitive-radio-inspired NOMA under varying channel gain disparities[cite: 4].
- **Findings:** Confirms that NOMA gain over OMA scales directly with channel state information (CSI) disparity between paired users[cite: 4].
- **Limitations & Project Relevance:** Does not cover multi-slot matching or learned policies[cite: 4]. Grounds the near-far baseline rationale.

[5] Ding, Peng, and Poor (2015) – Cooperative NOMA in 5G Systems

- **Problem & Method:** Investigates cooperative NOMA where strong channel users act as relays for weak channel users[cite: 5].
- **Findings:** Proves diversity gains for weak users without requiring dedicated relay infrastructure[cite: 5].
- **Limitations & Project Relevance:** Relaying is absent in the project repository. Included purely for channel-disparity context[cite: 5].

[6] Timotheou and Krikidis (2015) – Fairness for NOMA in 5G Systems

- **Problem & Method:** Formulates max-min fairness power allocation constraints in downlink NOMA[cite: 6].
- **Findings:** Solves optimal power coefficients to ensure equitable user data rates regardless of path loss[cite: 6].
- **Limitations & Project Relevance:** The project uses a fixed 0.75/0.25 power split; this paper motivates future dynamic power allocation extensions[cite: 6].

[7] Eisen and Ribeiro (2020) – Wireless Resource Allocation with Random Edge GNNs

- **Problem & Method:** Formulates resource allocation over wireless graphs using Random Edge GNNs (REGNNs) trained via primal-dual constrained optimization[cite: 7].
- **Findings:** Demonstrates scale-free generalization across varying topology sizes without manual heuristic retuning[cite: 7].

- **Limitations & Project Relevance:** Focuses on continuous power control rather than discrete pair/slot matching[cite: 7]. Strongest theoretical precedent for graph-structured resource management.

[8] Shen et al. (2021) – GNNs for Scalable Radio Resource Management

- **Problem & Method:** Establishes architectural guidelines (permutation equivariance/invariance) for GNN-based radio resource management[cite: 8].
- **Findings:** Proves theoretical conditions under which message passing networks match spatial optimization heuristics[cite: 8].
- **Limitations & Project Relevance:** General framework not specific to NOMA pairing[cite: 8]. Validates the choice of PyTorch geometric graph layers.

[9] Zhao et al. (2023) – Link Scheduling Using Graph Neural Networks

- **Problem & Method:** Maps link scheduling to maximum-weight independent set graph optimization using GNN scoring heads[cite: 9].
- **Findings:** Achieves near-optimal conflict-free link schedules with low inference runtime[cite: 9].
- **Limitations & Project Relevance:** Link scheduling differs from 2-user NOMA slot pairing, but the post-processing heuristic pipeline is identical in structure[cite: 9].

[10] Nikopour and Baligh (2013) – Sparse Code Multiple Access (SCMA)

- **Problem & Method:** Introduces multidimensional sparse codebook sets mapping user bits to shared resource elements via factor graph message passing[cite: 10].
- **Findings:** Shows shaping gains and overloaded access capacity over conventional CDMA/LMA[cite: 10].
- **Limitations & Project Relevance:** Standalone PHY proposal[cite: 10]. Basis for the project's separate 6-layer/4-resource MPA detector experiment.

[11] Yang, Liu, and Siu (2016) – Low-Complexity Message Passing for SCMA

- **Problem & Method:** Proposes a simplified dynamic search threshold MPA receiver to reduce dynamic exponential decoding complexity[cite: 11].
- **Findings:** Reduces dynamic floating-point operations by over 50% with negligible BER loss[cite: 11].
- **Limitations & Project Relevance:** Receiver-level optimization without scheduling logic[cite: 11]. Contextualizes the project's 4-iteration MPA engine.

[12] Nemalidinne et al. (2022) – DNN Pairing and Allocation with Imperfect SIC

- **Problem & Method:** Applies deep neural networks to approximate α-fair user pairing and power allocation under imperfect SIC.
- **Findings:** Demonstrates rapid execution speeds compared to non-convex optimization solvers.
- **Limitations & Project Relevance:** Employs fully-connected DNNs rather than topology-aware GNNs. Closest thematic precedent for imperfect SIC modeling.

4. Comparative Synthesis

Table 1 provides an academic mapping between literature benchmarks and the audited codebase.

Table 1: Comparative Synthesis: Published Literature vs. Project Implementation

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Ref.	Problem Domain	Published Findings	Project Mapping & Gaps
[1]	Downlink PD-NOMA PHY	Superior cell-edge sum rate via SIC[cite: 1]	Provides 2-user superposition model[cite: 1].
[2]	NOMA Overview / Challenges	Highlights dependency on CSI disparity[cite: 2]	Establishes baseline survey framing[cite: 2].
[3]	Optimal User Pairing	Proves pairing policy impact[cite: 3]	Precedent for near-far/oracle baselines[cite: 3].
[4]	Channel Disparity Pairing	Rate gains scale with CSI difference[cite: 4]	Grounds channel disparity heuristic[cite: 4].
[5]	Cooperative Relay NOMA	Achieves user diversity via relaying[cite: 5]	Not implemented; non-cooperative only[cite: 5].
[6]	Fair Power Allocation	Max-min power coefficient design[cite: 6]	Project fixes power split at 0.75/0.25[cite: 6].
[7]	Wireless Graph Optimization	Scale-free primal-dual REGNN[cite: 7]	Precedent for graph wireless control[cite: 7].
[8]	Scalable RRM Architecture	Permutation equivariant message passing[cite: 8]	Validates GNN layer formulation[cite: 8].
[9]	GNN Link Scheduling	Near-optimal conflict resolution[cite: 9]	Direct analogue to post-processing pipeline[cite: 9].
[10]	SCMA Codebook / MPA	Overloaded sparse access[cite: 10]	Basis for standalone MPA experiment[cite: 10].
[11]	Low-Complexity MPA	Reduced search tree decoding[cite: 11]	Contextualizes 4-iteration MPA engine[cite: 11].
[12]	DNN Allocation w/ Residual SIC	Deep learning approximation of allocation	Closest precedent; project adds GNNs.

5. Research Gaps and Project Motivation

1. **Integration Gap in Hybrid Scheduling:** The literature treats SCMA detection and NOMA pairing independently[cite: 1, 10]. While this project constructs a prototype covering both, SCMA evaluation in the main scheduler relies on a closed-form rate proxy rather than integrated MPA error performance[cite: 10, 11].

2. **Generalization & Oracle Dependence:** Training relies on exhaustive oracle labels. Evaluation across larger N or unseen spatial user distributions remains unverified in checked-in logs.

3. **Feature Sanity:** The scalar 'channel correlation' edge attribute evaluates to 1.0 uniformly for complex scalar channels, indicating a feature definition bug that should be rectified.

Recommended Project Framing

The project should be framed as: *"An implementation and empirical evaluation of an oracle-supervised, edge-conditioned GNN heuristic for joint user pairing and slot preferences within a synthetic hybrid-access simulator."* This phrasing accurately reflects the software artifacts without overstepping into unverified novelty claims.

References

[1] Y. Saito, Y. Kishiyama, A. Benjebbour, T. Nakamura, A. Li, and K. Higuchi, "Non-orthogonal multiple access (NOMA) for cellular future radio access," in *Proc. IEEE 77th Vehicular Technology Conference (VTC Spring)*, 2013, pp. 1–5.
DOI: 10.1109/VTCSpring.2013.6692652

[2] Z. Ding, X. Lei, G. K. Karagiannidis, R. Schober, J. Yuan, and V. K. Bhargava, "A survey on non-orthogonal multiple access for 5G networks: Research challenges and future trends," *IEEE Journal on Selected Areas in Communications*, vol. 35, no. 10, pp. 2181–2195, 2017.
DOI: 10.1109/JSAC.2017.2725519 — arXiv: 1706.05347

[3] L. Zhu, J. Zhang, Z. Xiao, X. Cao, and D. O. Wu, "Optimal user pairing for downlink non-orthogonal multiple access (NOMA)," *IEEE Wireless Communications Letters*, vol. 8, no. 2, pp. 328–331, 2019.
DOI: 10.1109/LWC.2018.2853741

[4] Z. Ding, P. Fan, and H. V. Poor, "Impact of user pairing on 5G non-orthogonal multiple access," *IEEE Transactions on Vehicular Technology*, vol. 65, no. 8, pp. 6010–6023, 2016.
DOI: 10.1109/TVT.2015.2480766 — arXiv: 1412.2799

[5] Z. Ding, M. Peng, and H. V. Poor, "Cooperative non-orthogonal multiple access in 5G systems," *IEEE Communications Letters*, vol. 19, no. 8, pp. 1462–1465, 2015.
DOI: 10.1109/LCOMM.2015.2441064 — arXiv: 1410.5846

[6] S. Timotheou and I. Krikidis, "Fairness for non-orthogonal multiple access in 5G systems," *IEEE Signal Processing Letters*, vol. 22, no. 10, pp. 1647–1651, 2015.
DOI: 10.1109/LSP.2015.2417119

[7] M. Eisen and A. Ribeiro, "Optimal wireless resource allocation with random edge graph neural networks," *IEEE Transactions on Signal Processing*, vol. 68, pp. 2977–2991, 2020.
DOI: 10.1109/TSP.2020.2988255 — arXiv: 1909.01865

[8] Y. Shen, Y. Shi, J. Zhang, and K. B. Letaief, "Graph neural networks for scalable radio resource management: Architecture design and theoretical analysis," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 1, pp. 101–115, 2021.
DOI: 10.1109/JSAC.2020.3036965 — arXiv: 2007.07632

[9] Z. Zhao, G. Verma, C. Rao, A. Swami, and S. Segarra, "Link scheduling using graph neural networks," *IEEE Transactions on Wireless Communications*, vol. 22, no. 6, pp. 3997–4012, 2023.
DOI: 10.1109/TWC.2022.3222781 — arXiv: 2109.05536

[10] H. Nikopour and H. Baligh, "Sparse code multiple access," in *Proc. IEEE 24th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2013, pp. 332–336.
DOI: 10.1109/PIMRC.2013.6666156

[11] L. Yang, Y. Liu, and Y.-M. Siu, "Low complexity message passing algorithm for SCMA system," *IEEE Communications Letters*, vol. 20, no. 12, pp. 2466–2469, 2016.
DOI: 10.1109/LCOMM.2016.2616082

[12] S. M. Nemalidinne, P. K. R. Manne, A. Kumar, and P. K. Upadhyay, "DNN based adaptive user pairing and power allocation to achieve α-fairness in NOMA systems with imperfections in SIC," in *Proc. 2nd International Conference on AI-ML Systems (AIMLSystems)*,

ACM, 2022, Art. no. 21.
DOI: 10.1145/3564121.3565042